

What is Transition in CSS?

-> The transition property in CSS is used to create smooth animation effects when a property changes from one value to another.

-> Without transition: Changes happen instantly.

-> With transition: Changes happen smoothly over time.

Transition Properties:

-> There are 4 main transition properties:

1. transition-property: For which property you have to give smooth behaviour

2. transition-duration: Duration for smooth change

3. transition-timing-function: Speed for the change

4. transition-delay: Provides delay before the smooth changes start

1. transition-property:

-> It is used to specify for which CSS property you want to apply smooth behavior (transition).

-> Syntax: transition-property: property-name;

-> Example:

```
.box {  
    transition-property: border-radius, color, etc;  
}
```

2. transition-duration:

-> It is used to specify the time duration for the smooth change to complete.

-> Syntax: transition-duration: time;

-> Time can be:

s = seconds

ms = milliseconds (1s = 1000ms)

-> Important: Without duration, transition will NOT work.

3. transition-timing-function:

-> It is used to specify the speed pattern/style of the smooth change.

-> Syntax: transition-timing-function: value;

-> Common timing functions values:

Value	Meaning
-----	-----
ease (default)	Slow start, fast middle, slow end
linear	Same speed throughout
ease-in	Slow start
ease-out	Slow end
ease-in-out	Slow start and end

4. transition-delay:

-> It is used to provide delay before the smooth change starts.

-> Syntax: transition-delay: time;

-> Example: transition-delay: 1s;

-> Transition starts after 1 second.

Shorthand Property:

-> Most commonly used.

-> Syntax: transition: property duration timing-function delay;

-> Example: transition: all 0.5s ease;

Here,

all -> Apply transition for all properties

0.5s -> Duration to complete the change

ease -> speed for the transition (slow->fast->slow) (timing-function)

Note: Some Properties That Cannot Transition Properly

1.display

2.position

3.float

Mistake to avoid:

-> Wrong✗

```
.box:hover {  
    transition: all 1s linear;  
}
```

-> Correct✓

```
.box {  
    transition: all 1s linear;  
}
```

-> Transition should be placed on the original element state.